VAKRATRON SYSTEMS

WHITEPAPER · PRIVATE AND HYBRID CLOUD

Private Cloud Without the Lock-in

Deciding what runs where, building a private cloud on OpenStack, KVM and Ceph, leaving VMware in planned waves, and running hybrid

Written for
CIOs, IT heads, infrastructure and cloud architects, finance partners

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What you will take away

- A five-question framework for deciding which workloads belong in private, public or hybrid cloud
- Private, public, hybrid and multi-cloud compared on cost, control and the skills each needs
- An OpenStack, KVM and Ceph reference design with starting sizes and the decisions that matter
- Compute, storage and five-year cost formulas with a worked 300-VM example, egress and licences included
- A wave-by-wave method for leaving VMware, with a rollback at every step
- Landing zones, upgrades and FinOps: running both clouds after go-live
- A 36-point checklist you can use with your team

01

Executive summary

Most organisations now run some workloads in a public cloud and some on infrastructure they control, and many are being pushed to rethink the second half by a VMware renewal. The question is no longer “cloud or not”. It is which workload runs where, at what five-year cost, and how to move without outages.

This paper covers placement, cost, a private cloud built on OpenStack, KVM and Ceph, a safe exit from VMware, and the landing zone and operating model that keep a hybrid estate under control.

Five things to take away

1. **Decide placement per workload, in writing.** Load shape, data movement, regulation, managed services and team skills decide where it belongs. A blanket “cloud first” or “all private” policy is usually the expensive answer.
2. **Compare five-year cost, not first-year cost.** Include people, data egress, licences, support and hardware refresh. Large, steady 24×7 estates often cost 25 to 40 percent less on a well-run private cloud; spiky or short-lived work is usually cheaper in public cloud.
3. **Private cloud means self-service, not just virtualisation.** OpenStack on KVM with Ceph gives teams an API and portal on hardware you own, with no hypervisor licence. It needs a small skilled team and a firm upgrade habit.
4. **Leave VMware in waves, with a way back.** Move dependency groups together, test side by side, and keep the source VM intact until the application owner signs off.
5. **Build the guardrails before the workloads.** One identity, a landing zone, central logging and mandatory tags take a few weeks up front, and far longer if retrofitted.

02

Why cloud programmes disappoint

Cloud programmes rarely fail with an outage. They disappoint slowly: the bill beats the business case, the private platform is feared, or the migration stalls at 60 percent. The causes are familiar.

What we find	What it leads to
Lift and shift with no right-sizing	VMs move as they were, sized for peak and running 24×7. The first annual bill is 1.5 to 2 times the estimate.
A renewal date driving the plan	The renewal quote arrives months before the date. The exit is rushed, or three more years are signed with no plan.
Accounts opened team by team	Nobody can say who has access to what, where the audit logs are, or whose costs are whose.
A private cloud nobody can run	Installed by a vendor and handed over without runbooks. Upgrades are postponed until support ends.
Data movement not costed	Applications that send large volumes out, or talk constantly across the link, cost more than their VMs.
Hybrid without design	Connectivity, DNS and identity were assumed rather than designed. Applications break in confusing ways after migration.

In short: most cloud problems are decisions that were never made explicitly: where each workload belongs, what it really costs, and who runs the platform afterwards.

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Placement, cost and lock-in, explained properly

What makes a private cloud a cloud

A virtualised data centre is not a private cloud. The difference is self-service: teams create their own VMs, networks and volumes through a portal or an API, within quotas, without raising a ticket. Projects are isolated, usage is metered, and everything can be built from code. If you only need to run VMs reliably and nobody will use an API, a simpler KVM platform such as Proxmox VE is often the better choice.

Workload shape decides cost

Public cloud charges for every hour a resource exists. A VM that runs a few hours a day is cheap there. A database running 24×7 at steady load pays the provider’s margin every hour for five years. Most enterprise estates are dominated by the second kind.

Data gravity and egress

Public clouds charge little to bring data in and charge per gigabyte to send it out. Large datasets also take days or weeks to copy. Keep data close to the applications that use it most, and cost every flow that crosses a cloud boundary.

Four kinds of lock-in

Kind	Example	How to limit it
Licence	Hypervisor subscription bundles, per-core database licences	Open hypervisor (KVM); check terms before placing databases
Data	Hundreds of terabytes in one provider's object storage	Keep a copy or an exit plan for critical data
Service	Applications built on a provider-only queue, database or AI service	Use managed services where the value is worth the dependence
Skills and tooling	Everything built by clicking in one console	Infrastructure as code; open monitoring and logging tools

> **Some lock-in is a fair price.** A managed database you would never want to run yourself is worth some dependence. Choose lock-in knowingly.

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The options compared

Most organisations end up using two or three of these models for different parts of the estate.

Model	What it means	Strong at	Watch out for
Private cloud	Self-service cloud on hardware you own or lease	Steady workloads, data that must stay put, predictable cost over 3 to 5 years	Needs a team that can run and upgrade it; capacity must be bought ahead of demand
Public cloud	AWS, Azure, OCI or similar, by the hour or on 1 to 3 year commitments	Spiky or uncertain load, fast start, managed databases, analytics and AI services	Cost grows quietly; egress and always-on VMs add up; governance must be built
Hybrid cloud	Private and public joined by private links, one identity and DNS	Keeping core data at home while using public cloud where it is stronger	Connectivity, identity and DNS must be designed, not assumed
Multi-cloud	More than one public cloud	A service only one provider offers well, or a customer requirement	Every extra cloud multiplies the guardrails, tools and skills to maintain

Choosing the private platform

For the private side, the right platform depends on how much self-service you need and what your team can operate.

Platform	Best for	Trade-off
OpenStack on KVM with Ceph	Multi-tenant, API-driven cloud for several teams; 100 to several thousand VMs	The most capable option, and the most to learn; needs a disciplined upgrade process
Proxmox VE	Running VMs reliably for one IT team, from a handful of hosts to a few dozen	Simple to run; limited multi-tenancy and self-service API
OpenShift Virtualization	Teams standardised on Kubernetes wanting VMs and containers together	Subscription cost; VM operations follow Kubernetes patterns
A small remaining vSphere island	Appliances and vendor applications only supported on VMware	Keeps a licence, so keep it small and review it every year

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Deciding what runs where

Decide placement per application, with a one-line reason agreed by its owner (“private: steady load, payment data must stay in India”). Five questions cover almost every case:

1. **What is the shape of the load?** Steady and 24×7 points to private. Seasonal, short-lived or fast-growing points to public.
2. **How much data moves, and where to?** If it sends many terabytes a month out, cost the egress first. Keep chatty applications next to the systems they talk to.
3. **What do regulators and contracts require?** Some data must stay in India, some on infrastructure you control, some needs audit rights over the provider.
4. **Does it rely on managed services?** An application built on a provider’s managed database or AI service belongs there, unless you will rebuild it.
5. **Who will run it?** A private platform needs Linux, networking and automation skills. Without them, public cloud or a managed private cloud is safer.

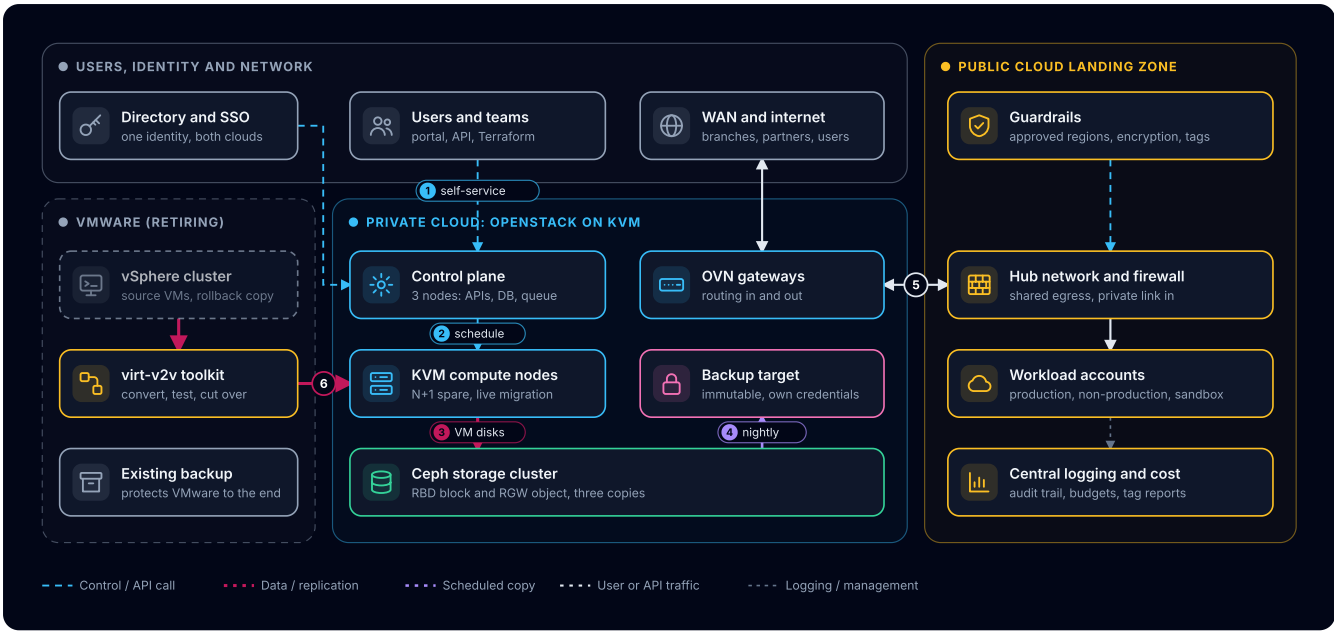
A typical estate comes out mixed:

Workload	Typical placement	Main reason
ERP, core banking, hospital information system	Private	Steady 24×7 load, regulated data, heavy integration
Customer website and mobile back end	Public, or hybrid with data private	Spiky traffic, content delivery networks, fast release cycles
Data warehouse and analytics	Depends on data volume	Managed analytics is attractive; moving large data out is not
Development and test	Public, scheduled off out of hours	Needed about 60 hours a week, not 168
Backup and long-term archive	Second site or public object storage	Cheap capacity, separate failure domain; cost the restore egress
GPU training bursts	Public or GPU cloud, then private for the steady base	Learn real usage before buying hardware

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Reference architecture

The diagram shows a hybrid estate during a VMware exit. A private cloud on OpenStack, KVM and Ceph runs the steady core; a public cloud landing zone runs what belongs there; both use one directory for sign-in. VMware shrinks until the last wave is signed off.



Numbered flows: (1) teams request VMs, networks and volumes through the portal, API or Terraform, signing in with their directory account; (2) the control plane places each VM on a compute node with room; (3) VM disks live on Ceph, not on the compute node, which makes live migration and fast recovery from a failed host possible; (4) backups go nightly to a separate, immutable target; (5) a private link such as Direct Connect, ExpressRoute or FastConnect joins the data centre to the cloud hub, with an IPsec VPN as backup; (6) VMs are converted from vSphere with virt-v2v and moved in waves.

Starting sizes for a first private cloud

Role	Starting point	Why
Control nodes	3	The database and queue need a majority to survive one node failing or being upgraded.
Compute nodes	Sized from inventory, plus 1 spare	Lets you patch or lose one host without running out of room.
Ceph storage nodes	3 minimum, 5 or more preferred	Three copies need three nodes; more nodes mean gentler recovery after a failure.
Networks	Management, storage, tenant overlay, external	Storage on its own network; 25 GbE to each server is a common baseline.

Decisions that make or break it

- **Choose the deployment tool for the next five years.** Kolla-Ansible, OpenStack-Ansible and commercial distributions all work, if your team can upgrade with them.
- **Design networks before installing anything.** VLAN ranges, overlay networks, external IP ranges, MTU and an IP plan that does not clash with the cloud are painful to change later.
- **Size Ceph for recovery, not just capacity.** When a node fails, Ceph rebuilds its copies onto the remaining nodes. Leave free space and network capacity for that to happen without hurting users.
- **Add the services teams expect.** Load balancing (Octavia), DNS (Designate), secrets (Barbican) and instance high availability (Masakari).
- **Decide who owns it.** A private cloud is a product, not a project. Someone must own upgrades, capacity and on-call from day one.

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Sizing and five-year cost

Compute

Size from the VM inventory, by processor and by memory, and take whichever needs more servers. Memory is rarely overcommitted; processor cores commonly are, at about three virtual CPUs per physical core.

```
Compute nodes = max( total vCPU ÷ (cores per node × overcommit × 0.75), total VM RAM ÷ (usable RAM per node × 0.85) ) + 1 spare
```

Storage

Ceph keeps three copies of each block, and must keep enough free space to rebuild a failed node's data onto the others before it reaches its 85 percent "near full" warning.

```
Usable capacity = raw capacity ÷ 3 copies × fill target, where fill target ≈ 0.85 × (storage nodes − 1) ÷ storage nodes
```

A worked example: 300 VMs

An estate of 300 VMs averages 4 vCPU, 16 GB of memory and 150 GB of disk each: 1,200 vCPU, 4,800 GB of memory and 45 TB of provisioned disk. Compute nodes have 64 physical cores and 768 GB of memory, of which 704 GB is usable after the host's own needs.

Item	Arithmetic	Result
Nodes by processor	1,200 ÷ (64 × 3 × 0.75 = 144)	8.3, so 9 nodes
Nodes by memory	4,800 ÷ (704 × 0.85 = 598)	8.0, so 9 nodes
Compute at go-live	9 + 1 spare	10 nodes, 12 by year three at 20% growth
Storage needed	45 TB provisioned + 30% growth	About 58 TB usable
Ceph fill target, 6 nodes	0.85 × 5 ÷ 6	About 0.71
Ceph raw capacity	6 nodes × 12 × 3.84 TB NVMe = 276 TB; 276 ÷ 3 × 0.71	About 65 TB usable

Five-year cost, both ways

The same 300 VMs over five years. Planning ranges at typical Indian prices in 2026, not quotations.

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Five-year cost = upfront hardware + expansion + 5 × (space and power + support and licences + people + backup, DR and links)
```

Cost line, five years	Private cloud (OpenStack, colocation)	Public cloud (3-year commitments)
Hardware: 3 control, 10 compute and 6 storage servers, network; 2 more compute nodes in year 3	₹3.5 to 5.5 crore	Included in hourly rates
Compute and storage consumption	Included above	₹13 to 18 crore
Space and power: about 4 racks	₹2.5 to 3.5 crore	Included
Software support, hardware maintenance	₹1.5 to 3 crore	₹0.7 to 1.5 crore (support plan)
Egress, NAT, load balancers, logging	Small	₹1.5 to 3 crore
Backup, DR and network links	₹1 to 2 crore	₹1 to 2 crore
People: platform or cloud operations team	₹4 to 5 crore (3 engineers)	₹2.5 to 3.5 crore (2 engineers plus FinOps)
Five-year total	**₹12.5 to 19 crore**	**₹18.7 to 28 crore**

On these assumptions the private cloud costs roughly a third less over five years. The gap narrows quickly if the private platform runs half empty, if the team must be hired at senior rates, or if right-sizing removes 20 to 30 percent of the public cloud capacity. Run the model with your own inventory.

Egress and licences: the lines most often missed

- **Egress:** at list prices of roughly ₹7 to 9 per GB, an application sending 20 TB a month out costs about 20,000 × ₹8 = ₹1.6 lakh a month, close to ₹20 lakh a year, before any VM is counted.
- **Windows Server and SQL Server:** per-core licences follow the physical cores of the host on any hypervisor. Check licence mobility rules before moving to public cloud.
- **Oracle Database:** licensing depends on how the hypervisor partitions processors; on many KVM platforms every host core can count. Check before a database moves.
- **Hypervisor and cloud software:** KVM, OpenStack and Ceph have no licence fee. A support subscription is optional but usually wise for production.

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Landing zones, security and governance

A landing zone is the set of accounts, guardrails, networks and logging in place before the first workload arrives. Built first, from code, it takes a few weeks. Retrofitted after two years of ad hoc accounts, it is a project of its own. On the private cloud, projects and quotas play the role of accounts.

Area	What good looks like
Identity	One sign-in through your directory, with roles in each cloud and in OpenStack. No local users or shared passwords; multi-factor everywhere; sealed break-glass accounts.
Account structure	Production, non-production and sandbox in separate accounts (projects on OpenStack), so a sandbox mistake cannot reach production.
Guardrails	Policies teams cannot switch off: approved regions, encryption and logging on, no public buckets, mandatory tags.
Network	Hub and spoke with a firewall and shared egress in the hub, and an IP plan that does not overlap with the data centre.
Logging	Audit logs and alerts from every account and cloud flow to one place teams cannot change.
Cost	Owner, environment and cost-centre tags enforced at creation; budgets and alerts per account.

Regulation in practice

Indian rules shape placement more than architecture. RBI requires payment system data to be stored in India and expects audit and access rights over outsourced IT, including cloud; IRDAI and SEBI set similar expectations. CERT-In requires incidents to be reported within six hours and logs kept for 180 days within India. The DPDP Act sets duties on how personal data is processed and secured. Residency is achievable in Indian public cloud regions or a private cloud; the real questions are audit rights, log retention and who holds the keys.

> **Keys and secrets:** encrypt volumes and object storage on both sides, keep keys for sensitive data in a service you control (customer-managed keys in public cloud, Barbican with a hardware security module on the private side), and rotate certificates on a schedule.

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Running it: day 2, upgrades and FinOps

Private clouds that disappoint were usually built well and then left without upgrades or capacity planning. Public cloud estates that disappoint never had an owner for cost. Both are fixed by a routine.

Activity	How often	What good looks like
Health and alert review	Daily	Alerts that matter reach a person. Noise is tuned out, not ignored.
Security patching	Monthly	Rolling host patching with live migration; no user downtime.
Capacity review	Monthly	Forecast of when compute, storage and IP ranges run out, with time to buy.
Restore test	Monthly	One real restore, timed and recorded.
Cost review (public cloud)	Monthly	Idle and oversized resources removed; commitments matched to steady use.
Platform upgrade	Once a year	Rehearsed on staging, rolled out with a written rollback plan.

Upgrades without fear

OpenStack releases every six months. Since the 2023.1 release, every other release supports an upgrade that skips one, so a yearly cycle is practical. Keep a small staging cloud built from the same code, rehearse each upgrade on it, and never fall more than one upgrade step behind. Platforms that skip upgrades for two or three years end up needing a migration instead.

FinOps for the public side

- **Right-size first.** Lifted VMs are often 30 to 50 percent larger than needed. Resize from four weeks of usage data before buying commitments.
- **Commit to the steady base only.** Reserved instances and savings plans cut 30 to 60 percent off compute for one or three years. Never commit to the peak.
- **Schedule non-production.** Off from 8 pm to 8 am and at weekends, development and test run 60 hours a week instead of 168, saving about 60 percent.
- **Show cost by team every month.** A report by owner tag changes behaviour faster than any policy.
- **Watch data transfer.** Egress, NAT gateway and cross-zone traffic are often the fastest-growing lines.

> **Tooling:** Prometheus, Alertmanager and Grafana for metrics, OpenSearch or Loki for logs, Ansible and Terraform for change, with the whole platform kept as code in version control.

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Roadmap: leaving VMware in planned waves

For a few hundred VMs, a private cloud build and VMware exit typically takes six to nine months. The pace is set by application owners' testing time, not conversion speed.

Stage	Typical duration	Outcome
1. Assess	3 to 4 weeks	Inventory (an RVTools export), dependency map, placement, five-year cost model, licence review.
2. Design	3 to 4 weeks	Architecture, network and IP plan, identity, bill of materials, wave plan, rollback rules.
3. Build and prove	6 to 10 weeks, after hardware arrives	Platform built from code and failure-tested; monitoring and backup working first.
4. Pilot wave	2 weeks	10 to 20 low-risk VMs covering every operating system version.
5. Production waves	8 to 16 weeks	Waves of 20 to 60 VMs every one or two weeks, each tested side by side before cutover.
6. Decommission	2 to 4 weeks	Last sign-offs, VMware hosts retired, licences dropped.

How a wave works

1. **Group by dependency.** An application's web, application and database VMs move together, so a cutover never splits a live dependency across a WAN link.
2. **Convert and test side by side.** virt-v2v copies each VM, replaces VMware drivers with virtio drivers and removes VMware Tools. Large VMs are pre-copied and synced in the window to keep cutover

short.

3. **Cut over only after sign-off.** The application owner tests on KVM while the VMware copy is still live. Users move only when the owner agrees.
4. **Keep the way back.** Source VMs stay intact and powered off for an agreed period, typically two to four weeks. Rollback is a DNS or IP change and a power-on.

What usually catches teams out

- **Drivers and boot.** Windows VMs need virtio drivers before or during conversion, or they will not boot. Test every operating system version in the pilot.
- **Network identity.** MAC addresses usually change. Anything licensed against a MAC address, and static DHCP reservations, need attention.
- **Vendor support.** Some appliances and commercial applications are supported only on VMware. Check before planning their wave; a few may stay on a small vSphere island or move to public cloud.
- **Features and storage.** Automatic load balancing like DRS is more limited on KVM, and Ceph behaves differently from VMFS and vSAN. Keep headroom and design volumes for the new platform.

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Ten common mistakes

1. **A blanket cloud policy.** “Cloud first” or “all private” puts some workloads in the wrong place by design.
2. **Comparing first-year cost.** Five years, with people, egress, licences and refresh, is the only fair comparison.
3. **Lifting VMs without right-sizing.** You pay every hour for capacity nobody uses.
4. **Letting the renewal date set the plan.** Start the exit assessment at least 12 months before the VMware renewal.
5. **Choosing OpenStack when Proxmox VE would do,** or the reverse. Pick the smallest platform that meets the need.
6. **Skipping upgrades.** A private cloud three releases behind needs a migration, not an upgrade.
7. **Converting VMs alphabetically.** Waves must follow dependencies, or a cutover splits a live application.
8. **Decommissioning VMware too early.** Keep the source VMs until the owner signs off, and the hosts until the last wave is done.
9. **Two identities.** Two audit trails and a clean-up project nobody planned.
10. **No owner for cost.** Without tags, budgets and a monthly report by team, public cloud cost only goes one way.

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Cloud readiness checklist (36 points)

Use this list to score your current position. Anything you cannot tick with evidence is a gap worth closing.

Strategy and placement

- [ ] Every application has a written placement and a one-line reason
- [ ] Placement agreed with each application owner
- [ ] Residency and audit requirements listed per application
- [ ] Dependencies mapped, so related systems are placed and moved together
- [ ] A named owner for the private platform and for public cloud cost
- [ ] Multi-cloud used only for a stated, specific need

Cost and commercial

- [ ] Five-year cost model built for private and public, from your own inventory and quotes
- [ ] Egress and cross-boundary data transfer estimated per application
- [ ] Database and operating system licences checked for the target platform
- [ ] VMware renewal date known, and the exit assessment started at least 12 months before
- [ ] Public cloud commitments limited to the steady base after right-sizing
- [ ] Hardware refresh and expansion included in the private cost

Platform design

- [ ] Platform choice justified against need and team skills
- [ ] Three control nodes and at least one spare compute node
- [ ] Ceph fill target stays below near-full after losing a node
- [ ] Separate networks for management, storage, tenant traffic and external access
- [ ] IP plan that does not overlap between data centre and cloud
- [ ] Deployment tool chosen for a five-year upgrade path, with a staging cloud

Migration

- [ ] Every VM assigned to a dependency-based wave
- [ ] Pilot wave covering every operating system version in use
- [ ] Monitoring and backup working on the target before the first production wave
- [ ] Rollback rule agreed: source VMs kept intact for a stated period
- [ ] Vendor support on KVM confirmed before each wave
- [ ] Owner sign-off recorded before each cutover

Security and governance

- [ ] Single sign-on from one directory to OpenStack and every public cloud account
- [ ] Separate accounts or projects for production, non-production and sandbox
- [ ] Guardrails for regions, encryption, logging, public access and tags
- [ ] Central audit logs kept for at least 180 days within India
- [ ] Keys for sensitive data held in a service you control
- [ ] Break-glass accounts sealed, monitored and tested

Operations and FinOps

- [ ] Runbooks for patching, upgrades, node replacement and certificate rotation
- [ ] Monthly capacity forecast for compute, storage and IP ranges
- [ ] Yearly upgrade rehearsed on staging before production
- [ ] Monthly cost report by team, with budgets and alerts
- [ ] Non-production environments scheduled off out of hours
- [ ] Monthly restore test, timed and recorded

13

Glossary

Term	Meaning
Private cloud	Self-service, multi-tenant infrastructure on hardware you own or lease, driven by a portal and API.
OpenStack	Open-source cloud software: Nova (compute), Neutron (network), Cinder (block storage), Keystone (identity).
KVM	The hypervisor built into Linux, used by OpenStack, Proxmox VE and most public clouds.
Ceph	Open-source distributed storage providing block (RBD), object (RGW) and file storage across many servers.
OVN	Open Virtual Network: the software-defined networking used by Neutron.
virt-v2v	An open-source tool that converts VMware VMs to run on KVM, swapping in virtio drivers.
Landing zone	The accounts, guardrails, network and logging set up before workloads arrive in a cloud.
Egress	Data sent out of a public cloud, charged per gigabyte.
FinOps	The practice of making cloud cost visible to the teams who create it, and managing it monthly.

About Vakratron Systems

Vakratron Systems LLP is a vendor-neutral infrastructure design firm, recognised as a startup by DPIIT. We design data centre, disaster recovery, cloud, GPU and AI platforms for enterprises and government buyers. Every design starts from the workload, compares realistic options with their costs, writes its assumptions down, and is tested before handover.

How we can help with private and hybrid cloud

- **Cloud assessment:** from your VM inventory export, a placement for every application, a five-year cost model and a migration order, in three to four weeks.
- **Private cloud design:** OpenStack, KVM and Ceph reference design, sizing, network plan, bill of materials and an upgrade plan your team can own.
- **VMware exit plan:** dependency-based waves, pilot wave, rollback rules and licence checks, with support through the first production waves.

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